Inquiry-Based Learning Task: The *Community Lift Design Challenge*

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Common Area of Study Across Two Course Pathways

Science education aims to develop students' ability to explain phenomena and solve problems using evidence and reasoning. This requires more than learning definitions or following fixed procedures. Students need regular opportunities to apply concepts to real systems. They need to interpret the results of their own investigations and communicate conclusions in clear language. Familiar and observable contexts make abstract ideas easier to test and discuss. Students can see the effect of their choices and compare their predictions with what actually happens. When a task can be adapted across more than one course, the instructor can keep the same core context. The instructor can adjust the complexity of thinking, the amount of mathematics, and the form of the final product. This supports coherence in the science program. It also helps align learning with different student pathways while returning to a shared set of big ideas. Within this framework, Grade 11 University Preparation Physics (SPH3U), and Grade 12 College preparation Physics (SPH3C), form a natural pairing. Both courses address the same core concepts, but they differ in the level of abstraction and mathematical demand (Ontario Ministry of Education, 2008). This makes them well suited to share a common context while still supporting pathway appropriate expectations.

Mechanical lifting systems fit this goal and provide a clear common topic for SPH3U and SPH4C. Elevators and lifts rely on the same physics for both courses. A lifting device must apply an upward force that overcomes the weight of a load. It must convert input effort into output torque at a drum or pullet. The system does work to increase the gravitational potential energy of the load, but some of the input energy is lost through friction and other inefficiencies (DiGiuseppe, 2011; Hirsch, 2003). These losses affect performance. The lift may move smoothly, stall, slip, or become unstable. This gives students clear evidence to analyze and explain. The context also connects physics to responsible design. Students must consider control of motion, structural stability, and safe operation for users. For these reasons, elevator and lift systems provide a shared topic that supports both pathways while allowing different levels of analysis in SPH3U and SPH4C.

Common Topic: Mechanical systems and energy efficiency in lifting loads
Courses: SPH3U & SPH4C

Curriculum Expectations by Course and Strand

Tabulated Overview of Course Strands and Curriculum Expectations

This section presents the tabulated strands and curriculum expectations used for the inquiry task in SPH3U and SPH4C. Table 1 lists the full overall and specific expectations exactly as they appear in the curriculum document, along with their stand placement. Table 2 provides a condensed summary that includes only the strand and the expectation codes for quick review and planning.

Table 1

SPH3U and SPH4C Curriculum Expectations Used in the Inquiry Task

Course	Strand	Expectations
SPH 3U	**Strand A**: Scientific Investigation and Career Exploration	A1. demonstrate scientific investigation skills (related to both inquiry and research) in four areas of skills. (initiating and planning, performing and recording, analysing and interpreting, and communicating). A1.5. conduct inquiries, controlling relevant variables, adapting or extending procedures as required, and using appropriate materials and equipment safely, accurately, and effectively, to collect observations and data A1.11. communicate ideas, plans, procedures, results, and conclusions orally, in writing, and/or in electronic presentations, using appropriate language and a variety of formats (e.g., data tables, laboratory reports, presentations, debates, simulations, models) A1.12. use appropriate numeric (e.g., SI and imperial units), symbolic, and graphic modes of representation for qualitative and quantitative data (e.g., vector diagrams, free-body diagrams, algebraic equations)

Strand C: Forces

C1. analyse and propose improvements to technologies that apply concepts related to dynamics and Newton's laws, and assess the technologies' social and environmental impact;

- C1.2 evaluate the impact on society and the environment of technologies that use the principles of force

C2. Investigate, in qualitative and quantitative terms, net force, acceleration, and mass, and solve related problems;

- C2.1 use appropriate terminology related to forces, including, but not limited to: *mass, time, speed, velocity, acceleration, friction, gravity, normal force,* and *free-body diagrams*
- C2.2 conduct an inquiry that applies Newton's laws to analyse, in qualitative and quantitative terms, the forces acting on an object, and use free-body diagrams to determine the net force and the acceleration of the object
- C2.4 analyse the relationships between acceleration and applied forces such as the force of gravity, normal force, force of friction, coefficient of static friction, and coefficient of kinetic friction, and solve related problems involving forces in one dimension, using free-body diagrams and algebraic equations (e.g., use a drag sled to find the coefficient of friction between two surfaces)
- C2.5 plan and conduct an inquiry to analyse the effect of forces acting on objects in one dimension, using vector diagrams, free-body diagrams, and Newton's laws

C3. demonstrate an understanding of the relationship between changes in velocity and unbalanced forces in one dimension.

- C3.3 state Newton's laws, and apply them, in qualitative terms, to explain the effect of forces acting on objects

	Strand D: Energy and Society	D2. Investigate energy transformations and the law of conservation of energy, and solve related problems; D2.1 use appropriate terminology related to energy transformations, including, but not limited to: *mechanical energy, gravitational potential energy, kinetic energy, work, power, fission, fusion, heat, heat capacity, temperature,* and *latent heat* D2.3 use the law of conservation of energy to solve problems in simple situations involving work, gravitational potential energy, kinetic energy, and thermal energy and its transfer (heat) D2.4 plan and conduct inquiries involving transformations between gravitational potential energy and kinetic energy (e.g., using a pendulum, a falling ball, an object rolling down a ramp) to test the law of conservation of energy D3 demonstrate an understanding of work, efficiency, power, gravitational potential energy, kinetic energy, nuclear energy, and thermal energy and its transfer (heat). D3.1. describe a variety of energy transfers and transformations, and explain them using the law of conservation of energy D3.5 describe, with reference to force and displacement along the line of force, the conditions that are required for work to be done
SPH 4C	**Strand A**: Scientific Investigation Skills and Career Exploration	A1. demonstrate scientific investigation skills (related to both inquiry and research) in four areas of skills (initiating and planning, performing and recording, analysing and interpreting, and communicating); A1.10. draw conclusions based on inquiry results and research findings, and justify their conclusions with reference to scientific knowledge A1.11. communicate ideas, plans, procedures, results, and conclusions orally, in writing, and/or in electronic presentations, using appropriate language and a variety of formats

Strand C: Mechanical Systems	C2. Investigate forces, torque, work, coefficients of friction, simple machines, and mechanical advantage, and interpret related data; C2.2 analyse, in qualitative and quantitative terms, the forces (e.g., gravitational, frictional, and normal forces; tension) acting on an object in one dimension, and describe the resulting motion of the object C2.6 investigate, in quantitative terms, common machines (e.g., a bicycle, a can opener, a piano) with respect to input and output forces and mechanical advantage C2.7 construct a simple or compound machine, and determine its mechanical advantage (e.g., a pulley, a mobile, a can crusher, a trebuchet) C3. demonstrate an understanding of concepts related to forces and mechanical advantage in relation to mechanical systems. C3.3. explain, with reference to force and displacement, the conditions necessary for work to be done

Strand E: Energy Transformations	E2. investigate energy transformations and the law of conservation of energy, and solve related problems; E2.1. use appropriate terminology related to energy and energy transformations, including, but not limited to: work, gravitational potential energy, kinetic energy, chemical energy, energy transformations, and efficiency E2.2 use the law of conservation of energy to solve problems involving gravitational potential energy, kinetic energy, and thermal energy E2.3 construct a simple device that makes use of energy transformations (e.g., a pendulum, a roller coaster), and use it to investigate transformations between gravitational potential energy and kinetic energy E2.4 design and construct a complex device that integrates energy transformations (e.g., a mousetrap vehicle, an "egg-drop" container, a wind turbine), and analyse its operation in qualitative and quantitative terms E2.5 investigate a simple energy transformation (e.g., the use of an elastic band to propel a miniature car), explain the power and output, and calculate the energy E3. demonstrate an understanding of diverse forms of energy, energy transformations, and efficiency; E3.2 explain the energy transformations in a system (e.g., a toy, an amusement park ride, a skydiver suspended from a parachute), using principles related to kinetic energy, gravitational potential energy, conservation of energy, and efficiency

Note. This table lists the relevant overall and specific expectations for SPH3U and SPH4C by course and strand. The expectation statements are reproduced verbatim from the Ontario curriculum document (Ontario Ministry of Education, 2008).

Table 2

Summary of SPH3U & SPH4C Curriculum Expectation Codes Used in the Task

Course	Strand and Expectations
SPH3U	**Strand A** ***A1***: A1.5, A1.11, A1.12 **Strand C**

	C1: C1.2 ***C2***: C2.1, C2.2, C2.4, C2.5 ***C3***: C3.3 **Strand D** ***D2***: D2.1, D2.3, D2.4 ***D3***: D3.1 D3.5
SPH4C	**Strand A** ***A1***: A1.10, A1.11 **Strand C** ***C2***: C2.2, C2.6, C2.7 ***C3***: C3.3 **Strand E** ***E2***: E2.1, E2.2, E2.3, E2.4, E2.5 ***E3***: E3.2

Note. This table summarizes the SPH3U and SPH4C overall and specific expectations used in the inquiry task by course, strand, and expectation code only. Full expectation wording is provided in Table 1.

Alignment of Strands and Expectations With the Common Topic

The *Community Lift Design Challenge* is organized around one mechanical system, a gear driven elevator that lifts a load vertically. This common topic connects SPH3U and SPH4C expectations because the same ideas determine performance in both courses: forces on the load, friction and structural stability in the mechanism, work supplied through rotational input, and energy transfer into gravitational potential energy. The lesson sequence moves from a baseline trial to redesign and retesting. This design makes the expectations visible through an iterative routine of planning, testing, interpreting, and communicating, rather than through isolated skill checks. The strands and expectations informing the task outlined in the following sections are derived from **Tables 1 & 2** and the Ontario Science curriculum for Grades 11 and 12 (Ontario Ministry of Education, 2008).

Scientific investigation skills structure how students work through the task. Students demonstrate A1 by selecting evidence that matches the goal, carrying out trials, organizing results, looking for patterns, and reporting conclusions. In SPH3U, A1.5 is developed through controlled inquiry. Groups keep the payload and target height constant, use a consistent testing method, and change one feature at a time during redesign, such as gear ratio, spool size, axle alignment, or the routing and attachment of the lifting line. When results are inconsistent, students refine the procedure in ways that improve reliability while keeping comparison conditions stable, for example by repeating trials, tightening supports, reducing rubbing points, or clarifying measurement steps. SPH3U A1.11 is developed through continuous communication. Students record procedures and results in data tables, annotate labelled sketches, write brief explanations of design choices, and share findings during checkpoints and final

reporting. SPH3U A1.12 strengthens both analysis and communication because students represent evidence using appropriate units for mass, distance, and time, and they select representations that fit the question, including graphs for comparing configurations, algebraic relationships for linking variables, and free body and vector diagrams to support force based explanations. The same inquiry cycle supports SPH4C Strand A, with a stronger emphasis on justification. A1.11 is evident in consistent reporting of procedures and findings, while A1.10 is evident when students state what improved, support the claim with trial evidence, and connect the improvement to scientific ideas that are central to the task, including mechanical advantage, friction losses, and work.

The collected evidence then becomes the basis for force based explanation, since the lift outcome depends on whether the mechanism can overcome gravity and resist friction. In SPH3U, Strand C is developed by treating the payload motion as a one dimensional case where net force determines whether the load accelerates, moves at constant velocity, or remains at rest. Students demonstrate C2.1 through correct use of terms such as mass, time, speed, velocity, acceleration, friction, gravity, normal force, and free body diagrams to describe system behaviour. C2.2 is evident when students apply Newton's laws to analyse the forces on the payload, construct a free body diagram, and use it to determine the direction of the net force and the expected acceleration, especially when comparing a design that stalls with one that lifts smoothly. C2.5 is evident when students plan follow up tests that isolate how a design change alters the forces transmitted to the load, then interpret motion using vector reasoning, free body diagrams, and Newton's laws. C2.4 is addressed when students connect changes in motion to applied forces, including friction, and use diagrams and algebraic relationships to explain one dimensional situations that arise in their tests. Within the task, friction is treated as an opposing effect that reduces the net lifting effect, and students use start-up difficulty, slipping, and mid lift slowing as evidence that friction and misalignment are limiting performance. C3 and C3.3 are demonstrated when students state Newton's laws and use them qualitatively to explain how unbalanced forces change velocity during start up, steady motion, and slowing. This same evidence base also supports the technology expectations. The overall expectation of C1 is demonstrated through the redesign cycle, as students analyse the elevator as a force based technology and use test results to propose improvements. C1.2 is demonstrated when students evaluate impacts using performance criteria drawn from the task, including reliability, safety, required effort, and efficiency. In SPH4C, the same force ideas are interpreted through mechanical systems and mechanical advantage. Students demonstrate C2.2 by analysing forces acting on the load in one dimension, including gravitational, frictional, and normal forces and tension, and describing the resulting motion. They demonstrate the overall expectation of C2 by investigating forces, torque, work, coefficients of friction, simple machines, and mechanical advantage, then interpreting trial data to guide design decisions. Mechanical advantage is treated as a measurable property of the system rather than a label. Students demonstrate C2.6 and C2.7 when they compare input and output performance across configurations and determine mechanical advantage using evidence, such as the effort required and the input motion needed to

produce the same lift. C3 and C3.3 are developed when students connect force and displacement to explain when work is done and why changing the mechanism changes the relationship between input effort and input distance.

Expectations related to the energy strands are applied through what students can measure and explain in the lift systems, with a focus on mechanical energy transfers rather than a separate treatment of heat. In SPH3U, Strand D is developed as students use conservation of energy and the definition of work to interpret performance differences between designs. D2.1 is supported through accurate use of task relevant terminology, including work, gravitational potential energy, kinetic energy, power, and efficiency, with mechanical energy used as an umbrella term when describing the system. D2.3 is developed when students use conservation of energy to relate input work to the increase in gravitational potential energy of the payload, and to explain why designs that "feel harder" are not creating more gravitational potential energy, but are losing more of the input through friction and mechanical inefficiencies. D2.4 is developed when students plan and conduct inquiries that test conservation reasoning in a practical way by comparing predicted energy changes for the lift to evidence from baseline and redesign trials, then explaining discrepancies using design features such as friction points, slipping, or deformation. D3.1 is evident when students describe the energy pathway from input work at the handle, through the gear train, into the lifting line, and into the payload, with losses described in task appropriate terms as energy dissipated by friction and imperfect transmission. D3.5 is developed when students connect work to force and displacement along the line of force at both the input and output stages of the system. In SPH4C, Strand E frames the same core ideas through energy transformations and efficiency in devices. E2.1 is supported through correct terminology directly used in the lesson, including work, gravitational potential energy, kinetic energy, energy transformations, and efficiency. E2.2 is developed when students apply conservation of energy to explain the relationship between input work and gravitational potential energy gained, and to account for why different designs require different input even when the payload and height are unchanged. E2.3 is evident because students construct and test a device where gravitational potential energy and kinetic energy change during lifting. E2.4 is evident because the gear driven elevator functions as a multi component device that students analyse using qualitative observations and quantitative trial data. E2.5 can be incorporated when students isolate a specific efficiency limiting feature, explain how it affects useful output and power in practical terms, and calculate energy changes from measured quantities. E3 and E3.2 are developed when students explain the system's energy transformations using conservation and efficiency reasoning and support those explanations with results from baseline and redesign trials.

Student Learning Strengths and Learning Needs by Course Level

In Ontario, Grade 11 and 12 courses are divided into destination-related types: university preparation (U), university / college preparation (M), college preparation (C), and workplace preparation (E). University preparation courses are designed to equip students with the

knowledge and skills needed to meet university entrance requirements, while college programs, apprenticeships, or other training programs. University preparation courses are designed to equip students with the knowledge and skills needed to meet university entrance requirements, while college preparation courses are designed to support entry into college programs, apprenticeships, or other training pathways (Ontario Ministry of Education, 2008). These destination choices reflect students' postsecondary goals and shape how course content is emphasized and assessed, including the balance between conceptual depth, mathematical reasoning, and practical application. In physics, this can influence how students approach problem solving, interpret evidence, and connect models to real systems as they work through the learning task. The section that follows outlines the learning strengths and learning needs commonly observed in SPH3U and SPH4C learners and explains how these patterns may affect engagement and performance throughout the task.

Typical Learning Strengths of University-Streamed (Academic) Students

Students in the university preparation pathway generally bring the following strengths with them into the classroom:

- **Strong Academic Foundation from SNC2D:** SPH3U students have completed Grade 10 academic science (SNC2D) as a prerequisite. This means they have already been held to expectations that emphasize theoretical understanding, abstract reasoning, and multi-step problem solving. They arrive with experience analyzing the interrelationships between science, technology, society, and the environment at a conceptual level, not just an applied one.
- **Solid algebraic and equation manipulation skills:** Most SPH3U students are concurrently enrolled in, or have completed, academic mathematics courses that emphasize symbolic work and algebraic reasoning. They can usually rearrange formulas, isolate variables, and substitute numerical values with reasonable accuracy, allowing instruction to focus on physics concepts rather than foundational algebra review.
- **Relatively strong reading and scientific writing skills:** Academic pathway courses require students to interpret longer, more complex texts and to respond to questions using structured written explanations. SPH3U students are typically able to unpack multi-step word problems and express their scientific reasoning in complete, coherent written responses.
- **Developing independence and self-regulation as learners:** The academic stream places higher expectations on homework completion, self-study, and organization. Many SPH3U students can manage note-taking, seek clarification independently, and engage with textbooks or digital resources without constant teacher prompting.
- **Academic Motivation:** Students often select SPH3U with the explicit intention of pursuing university programs in engineering, science, health, or related fields. This goal orientation can increase their willingness to persist with challenging topics and to view the course as a critical step in their post secondary pathway.

- **Willingness to engage with abstract ideas:** Many SPH3U students are comfortable working with idealized models, such as frictionless surfaces, point masses, or constant acceleration. They may be more willing to think beyond everyday appearances in order to work with simplified theoretical models that support deeper understanding.
- **Capacity for independent study:** These students can learn from textbook sections, videos, or other resources outside of class time and come prepared with questions. They are capable of reviewing past content, practising additional problems, and seeking clarification when needed.
- **Higher tolerance for quantitative load:** Students can handle questions that involve multiple calculations, unit conversions, and the integration of more than one concept (for example, combining forces and energy or motion and graphs) within a single problem. They are generally comfortable working with several equations in the same solution.
- **Confidence in their academic abilities:** Students in the academic / university pathway often have had success with math and science courses, which contributes to a sense of confidence when approaching new physics topics. This confidence can support their willingness to attempt challenging problems and participate in class discussions.
- **Responsibility and organization:** Many students in the university stream demonstrate consistent homework completion, note-taking, and preparation for assessments. They can keep track of assignments and unit expectations with moderate teacher support.
- **Following and completing multi-step problems**: These learners can often follow a sequence of steps that involves identifying knowns and unknowns, selecting appropriate equations, performing calculations, and checking the reasonableness of an answer. They can handle questions that integrate several sub-parts into one larger problem.
- **Justifying conclusions using evidence and reasoning**: Students can support their answers using data from experiments, patterns in graphs, and logical arguments based on physical principles. They are able to move beyond simply giving a numerical answer to explaining why that answer makes sense.
- **Reading comprehension and processing of complex questions**: SPH3U students tend to manage multi-sentence questions that include background context, multiple conditions, and precise wording. They can extract relevant information, ignore distractions, and identify what is being asked.

Typical Learning Strengths of College-Stream (Applied/College Prep) Students

Students in college preparation pathway will bring distinct and valuable strengths with them into the classroom such as:

- **Strong applied thinking:** Students engage well when tasks are hands-on and connected to real-world contexts such as vehicles, lifts, tools, and everyday technologies. They are able to use these familiar situations to make sense of forces, motion, and energy.
- **Real-world and hands-on orientation:** They respond positively when physics is embedded in real scenarios (for example, cars, elevators, trade applications, safety

situations, and energy efficiency problems). In these contexts, they can reason effectively and make sensible, practical decisions.

- **Practical problem-solving skills:** Many students show strength in troubleshooting and improving physical systems. They are often comfortable using tools and equipment, noticing when something is not working as intended, and suggesting realistic adjustments.
- **Collaboration and oral communication:** Students frequently work well with classmates and explain ideas clearly in plain language during discussions and group work. When tasks feel relevant, they participate actively, share strategies, and support peers' understanding.
- **Drawing on lived experience:** Students may use personal and work experiences to interpret physics concepts, such as linking motion and forces to driving, lifting, or equipment use. This helps them connect abstract ideas to situations they already understand.
- **Resilience and adaptability:** Students who have progressed through applied and college pathways into Grade 12 have often developed perseverance and coping strategies. When they feel supported and see the relevance of a task, they can persist with challenging material.
- **Career-connected thinking:** Many students have specific college, apprenticeship, or workplace goals and regularly ask "Why does this matter for my future?" They engage more deeply when they can see how physics links directly to the skills and knowledge needed in those pathways.
- **Trade and technical skill awareness:** Given their pathway orientation toward college and workplace destinations, these students may already have exposure to skilled trades, technology, and applied sciences, providing rich prior knowledge for courses like SPH4C that cover a variety of mechanical systems and their components.

Typical Learning Needs of University-Stream Students

Despite their strengths, Grade 11 University students typically require:

- **Developing a physics-specific way of thinking:** For many students, SPH3U is their first full physics course; previous exposure occurred as a single unit within the general science courses. They require explicit support to adopt habits of mind such as consistently drawing diagrams, identifying known and unknown quantities, and justifying the choice of laws or equations used in problem solving.
- **Deepening understanding of vectors and direction:** Although students may have prior experience with direction in everyday contexts, they often lack a rigorous understanding of vector quantities. They need structured instruction and practice in representing vectors graphically, resolving them into components, and reasoning about motion and forces in one and two dimensions.

- **Interpreting graphs and mathematical relationships in physical context:** Although students may have prior success with graphing and representations in mathematics, they often struggle to interpret physics representations accurately. Common challenges include confusing the shape of a graph with an object's path and misreading axes or units. They need targeted instruction in interpreting slopes, areas, and intercepts on motion and other physics graphs or representations, and in linking mathematical relationships to physical meaning.
- **Connecting mathematical procedures to physical meaning:** Many students can substitute numbers into formulas and obtain a numerical result, but struggle to interpret what that result implies about the physical situation. They need learning experiences that explicitly link formulas and graphs to real motion, forces, and energy transfers, and that require them to explain results in conceptual terms.
- **Strengthening strategies for multi step, non-routine problems:** In earlier grades, problems often closely resembled textbook examples and involved one or two steps. SPH3U problems frequently demand choosing among several relationships, combining multiple steps, and checking the reasonableness of answers. Students benefit from structured problem-solving frameworks and frequent modelling of expert reasoning.
- **Addressing persistent physics misconceptions:** Common intuitive ideas, such as "heavier objects fall faster" or "a constant force is required to keep an object moving," often persist despite prior instruction. Learners need deliberate opportunities to surface these misconceptions, test them through demonstration or investigation, and reconstruct their understanding based on evidence.
- **Building scientific investigation and communication skills:** Even high achieving students may need explicit instruction in scientific investigation skills expected in SPH3U, including planning investigations, collecting and recording data, analyzing and interpreting results, and communicating findings. They benefit from practice discussing sources of error, critiquing models, comparing theoretical predictions with experimental results, and validating conclusions using evidence rather than equations alone.
- **Managing the intensity and pace of the course:** SPH3U covers several major strands, including kinematics, forces, energy, waves and sound, and electricity and magnetism, within a single semester. Students may need explicit support in organizing their notes, prioritizing key skills and concepts, and planning regular review to keep pace with the curriculum.
- **Building resilience and comfort with productive struggle:** Many SPH3U students have historically achieved higher grades (>75%) with comparatively less effort. When physics presents genuine difficulty, they may interpret this as a lack of ability rather than a normal part of learning. They require a classroom environment that normalizes challenge, emphasizes growth over perfection, and values reasoning as much as final answers.
- **Refining graphing and representation skills in a physics context:** Although they have graphed functions in mathematics, students often misinterpret physics graphs, confusing

the shape of a graph with the physical path or misreading axes. They need targeted instruction to interpret slopes, areas, and intercepts on motion graphs and to translate consistently between verbal descriptions, diagrams, graphs, and equations.

- **Connecting abstract concepts to real-world applications:** While students may be more comfortable with abstract models, they sometimes struggle to see how these ideas apply to everyday contexts. They need opportunities to link equations and diagrams to familiar situations, such as vehicles, sports, or everyday motion, so that physics feels meaningful and not purely theoretical.
- **Translating solutions into meaningful explanations:** Many students can complete calculations correctly but provide limited written or verbal explanations of what their answers imply. They need support in turning numerical results into clear statements about an object's motion, the size of a force, or the change in energy in a system.
- **Confidence with open-ended tasks that do not have a single "correct" answer:** SPH3U students are often accustomed to well-defined questions with one correct numerical solution. They may feel uncertain when faced with open-ended investigations, design challenges, or questions that ask for multiple reasonable approaches. They need guidance and reassurance that reasoning, justification, and creativity are as important as a single final answer.
- **Scientific communication and validation beyond equations:** Students sometimes equate "doing physics" with "getting the right number from the right formula." They need practice in other scientific practices, such as articulating assumptions, discussing sources of error, critiquing models, and comparing theoretical predictions with experimental results.
- **Reflecting on societal and environmental impacts:** While they can work effectively with technical content, Grade 11U students may give less attention to the broader implications of physics, such as safety, ethics, sustainability, and equity. They need structured tasks that prompt them to consider how physics-based technologies affect communities and the environment.
- **Collaborating effectively during group work:** Although many students can work near peers, not all are skilled at collaborative practices such as sharing roles, listening to alternative ideas, or building on others' reasoning. They need explicit expectations and support for productive group work in labs and problem-solving tasks.
- **Managing the volume and pace of content:** SPH3U covers several major strands within a semester, and the pace can feel demanding. Students may need help prioritizing key concepts, organizing their notes by unit, and building regular review habits so that concepts from earlier in the course remain accessible later on.
- **Balancing confidence with willingness to struggle:** Because many SPH3U students are used to achieving high marks, they can become discouraged when they encounter sustained difficulty. They need a classroom culture that normalizes challenge, frames

mistakes as part of learning, and provides structures (like re-tries, conferencing, and scaffolded tasks) that help them remain engaged when the work is hard.

- **Growth mindset and perseverance:** Physics from a conceptual perspective is demanding and often the first course where students encounter significant struggle. Building a growth mindset and normalizing productive struggle is an important learning need for the students.
- **Metacognitive and self-regulation development:** Although Grade 11 students are often expected to work independently, many are still developing metacognitive skills, including the ability to self-assess, set goals, and monitor their own understanding. As a result, explicit instruction in self-regulation strategies is beneficial.
- **Strengthening scientific investigation skills:** Although Grade 11U students may have prior success in science courses such as SNC2D, many still need explicit support in developing the full range of scientific investigation skills expected in SPH3U. This includes initiating and planning investigations, performing procedures and recording observations accurately, analysing and interpreting data, and communicating conclusions clearly using appropriate scientific language and evidence.

Typical Learning Needs of College-Stream Students

Grade 12 College preparation physics students typically require:

- **Consolidating foundational numeracy and algebra skills:** Because SPH4C admits students from both academic (SNC2D) and applied (SNC2P) backgrounds, there is often wide variation in mathematical readiness. Many learners require explicit review of operations with fractions and ratios, solving simple equations, and working with scientific notation and significant figures, not only in an initial "review unit" but throughout the course.
- **Rebuilding confidence in mathematics and science:** Students who have gone through the applied pathway may have received lower marks and negative messages about their capabilities. They may be quick to label themselves as weak in math or science and disengage early when a task appears difficult. They need structured opportunities for success, targeted feedback, and an explicit focus on their existing strengths.
- **Strengthening reading comprehension for technical tasks:** Multi-step word problems, dense laboratory instructions, and specific vocabulary can present significant barriers to understanding. Students may miss important details or misunderstand the requirements of a question. They benefit from chunked instructions, visuals embedded with text, opportunities to paraphrase tasks, and explicit vocabulary teaching.
- **Scaffolding for mathematics used in physics:** Students may need structured support with the math embedded in physics, including rearranging equations, substituting values correctly, interpreting graphs, and using basic trigonometry where required. Frequent, short math reviews within lessons help maintain access to the physics content.

- **Frequent feedback and practice to build confidence:** Students benefit from multiple low-stakes practice opportunities with timely feedback. This allows them to correct misunderstandings early, experience success, and gradually build confidence with quantitative and conceptual tasks.
- **Resilience after errors or setbacks:** When students get a question wrong, some may disengage or shut down. They need a classroom culture that normalizes mistakes, emphasizes learning from errors, and treats revisions and retries as part of the learning process.
- **Sustained attention to multi-step processes**:Even though the problems are less algebraically complex than in university-stream physics, many tasks still require carrying out several steps in order. Students benefit from clear problem-solving templates, checklists, and modelling of how to move through each step without losing track.
- **Connecting abstract concepts to concrete experience:** Students often understand concrete experiences well but may struggle to link them to abstract representations and formulas. They need deliberate bridges from hands-on activity to diagrams, graphs, and symbolic relationships that show how their experiences match the underlying physics.
- **Graph interpretation and data analysis:** Many students require support to read and create graphs, identify trends, and interpret slopes or changes in data. Explicit modelling of graphing from experimental data and guided questions about "what the graph is telling us" help deepen this skill.
- **Organization, time management, and study skills:** Some students still need support to keep notes organized, track assignments, and prepare effectively for assessments. Visual timelines, structured note templates, and explicit instruction on how to review for tests can improve their ability to manage the course demands.
- **Engagement with difficult or unfamiliar topics:** When students cannot see a clear reason for learning a topic, they may ask "Why do I need this?" and disengage. They need frequent, explicit explanations of relevance, along with examples and tasks tied to trades, college programs, and real-world problems that make the value of the learning visible.

Information for this section was derived from a comparison of the overall and specific expectations for SPH3U and SPH4C in the curriculum document (Ontario Ministry of Education, 2008). This comparison was supplemented with sources describing destination related pathways and typical learner profiles in university and college preparation courses. (Halton Catholic District School Board, 2020; Ontario Virtual School, 2025; Canadian Virtual School, n.d.; People for Education, 2020).

Inquiry-Based Learning Task Design and Implementation

Introduction

Science as a discipline advances through cycles of questioning, investigation, and revision in light of new evidence (Windschitl et al., 2008). This dynamic and iterative process has driven technological innovation, medical breakthroughs, and deeper understanding of the natural world. Traditional teacher-centered models of instruction do not fully reflect this authentic nature of science or how scientific knowledge is produced. In these models, the teacher is often positioned as the primary source of knowledge, and students are expected to receive and memorize information. Concepts may also be explained in a single fixed way, which may not align with the diverse prior knowledge, ways of thinking, and the learning needs of all students. As a result, some students may not develop deep conceptual understanding or transferable problem-solving skills. In contrast, student-centered approaches position learners as active participants who construct understanding through engagement with meaningful problems, collaboration, and reflection (Freeman et al., 2014).

In science education, one way to support this kind of learning is through inquiry-based learning. Inquiry-based learning can be defined as a pedagogy in which students investigate questions, problems or phenomena by collecting and analyzing evidence, generating and testing explanations, and communicating and defending their ideas, rather than simply reproducing information provided by the teacher (Pedaste et al., 2015). Within this approach, inquiry can be differentiated by the degree of students' autonomy and can be understood as a continuum from more teacher-directed to more student-direct experiences. This includes structured or controlled inquiry, in which the teacher provides the question and procedure and students carry out the investigation and interpret results. It also includes guided inquiry, in which the teacher poses the question but students design the methods and make decisions about how to collect and analyze data, and open inquiry, in which students formulate testable questions, design and carry out investigations, and develop and justify their own explanations (Zion & Mendelovici, 2012). To support teachers in implementing these forms of inquiry in a coherent and intentional way, the 5E instructional model (Engage, Explore, Explain, Elaborate, and Evaluate) offers a research-informed structure for sequencing learning experiences (Cheng et al., 2016; Duran & Duran, 2004). Through this structure, students' questions are elicited, investigations are conducted, evidence-based explanations are developed, and understanding is extended and assessed (Duran & Duran, 2004). This makes the 5E model a practical framework for implementing inquiry-based, student-centered learning in high school science and physics classrooms.

Outline of the Inquiry-Based Learning Task

Building on this understanding of inquiry-based, student-centered learning, the task is designed as a multi-day inquiry activity in which students design, test, and refine a gear-driven elevator system that can lift a load vertically. The objective is for students to apply ideas about

forces, torque, gear ratios, mechanical advantage, friction, and stability to improve the performance of a mechanical system through iterative, evidence-based design. Rather than following a fixed laboratory structure, students make design decisions, test their ideas under consistent conditions, and use their own observations and measurements to guide further modifications. In this way, the task models how scientific and engineering knowledge develops through cycles of trial, reflection, and revision. The task focuses on a specific set of physics concepts, using the elevator system as the shared context for learning. Students work with gear ratios and explore how changing the relative sizes and arrangements of gears affects output torque, speed of motion, and the ease of lifting a load. They investigate mechanical advantages by comparing systems that lift the same load with different amounts of effort or at different speeds, and they consider the trade-offs that arise when a design becomes easier to turn but slower, or faster but harder to control. As they build and test their systems, students also confront the practical effects of friction, misalignment, and structural stability. Signs of friction (such as jerky motion, squeaking, or stalling), instability (such as tipping or wobbling), and inconsistent performance become starting points for questions about how to redesign the system. Through these experiences, students experience concepts such as torque, mechanical advantage, and energy loss directly, rather than treating them as isolated formulas (DiGiuseppe, 2011; Science North, n.d.).

Implementation of the task spans four class periods, each contributing to a distinct stage in the overall sequence. On the first day, students work in small groups to construct an initial gear-driven elevator using common materials such as cardboard or foam board for the frame, gears of various sizes, dowels, or skewers as axles, string or yarn for the cable, tape, masses, and a timing device (Science North, n.d.). They are encouraged to experiment with different gear configurations, spool diameters, axle placements, cable paths, and supports. The primary aim is simply to produce a functioning preliminary design and test it under consistent conditions. During these baselines trials, students record qualitative observations related to ease of lifting, speed, smoothness, stability, and evidence of friction. In a short consolidation, each group identifies specific weaknesses in its design (for example, requiring too much effort, moving too slowly, shaking or failing under load) and proposes potential modifications that might address those weaknesses. This day introduces the design problem, surfaces prior ideas about forces and simple machines, and generates questions about how to improve the system.

The second day moves from exploratory building toward more structured comparison of designs. Each group implements one intentional design change based on the limitation identified in its baseline trial. Examples include changing to a different gear ratio to increase torque, altering the spool size to change the distance per turn, adjusting axle alignment to reduce friction, or reinforcing supports to improve stability. Before testing, the class revisits the idea of a fair comparison and agrees on which conditions must remain constant, such as the mass being lifted and the basic testing procedure. Groups then re-test their elevators using the same conditions as before and record new observations and simple measurements, such as time to lift the load or number of turns required. By comparing their first and second trials, students begin to evaluate

whether the modification improved performance, created trade-offs, or introduced new problems. This day makes the link between design choices and system behavior more explicit and encourages students to reason about cause and effect in their designs.

On the third day, the emphasis shifts from additional building to organizing and explaining results. Students draw on their baseline and modified trials to construct clear statements about how their elevator performs and how their design changes affected it. They organize their data and observations into tables, simple graphs, and annotated diagrams that show the configuration of gears and other key components. As they prepare these representations, students are prompted to connect their results to underlying physics ideas, such as torque (how changing lever arm or gear ratio affected the effort required), mechanical advantage (how the system trades off force and distance), and sources of energy loss (how friction or misalignment reduced efficiency). The lesson is framed as preparation for a design "pitch," so groups decide how best to present their central claim, which pieces of evidence are most convincing, and which visuals will help an audience quickly understand their system. This stage helps students move from raw results to clear, evidence based explanations.

The fourth day culminates the inquiry sequence in a set of structured group presentations and reflective discussion. Each group presents their final elevator design, explaining the initial design problem it faced, the key design decisions it made, the main modification or modifications implemented, and the evidence that performance changed in a meaningful way. Presentations include a brief overview of the system, specific description of the design change (for example, replacing a small driven gear with a larger one to increase torque), and before-and-after comparisons using the data and observations they collected. Groups explain the trade-offs created by their choices and relate these trade-offs to the physics concepts they have been working with, such as why a design that is easier to turn may move more slowly or why reducing friction can improve both speed and reliability. Questioning from peers and the teacher further presses students to justify whether their comparisons were fair and whether their evidence supports their claims. Through this final step, students practise sharing and defending design decisions through evidence based discussion and feedback.

This task also supports differentiation and can function as a low-floor, high-ceiling activity that is accessible to a wide range of learners. At the entry level, all students can participate in constructing a basic elevator system, making simple observations about how it behaves, and describing obvious problems such as needing too much effort or moving unevenly. Students who are less comfortable with mathematics can focus on qualitative comparisons, clear descriptions of what changed, and practical reasoning about why a design works better in everyday terms. At the same time, the task offers extension opportunities for students who are ready to engage more deeply with the physics, such as reasoning about torque and mechanical advantage in more detail, using numerical data to compare designs, or considering energy transfer and efficiency. Because the core materials and structure remain the same, the same task can be used with both grades with only minor adjustments to the level of formality and depth of

explanation expected, allowing the next section of the paper to address these adaptations explicitly.

Integration of Inquiry-Based Learning in the Task

The *Community Lift Design Challenge* is structured as an inquiry based learning task because students are not following a prescribed set of steps to confirm a known outcome, rather, they are engaging in a cyclical process of questioning, investigating, redesigning, explaining, and defending evidence based on conclusions. Inquiry is incorporated through the deliberate position of students as engineers and decision makers who must determine what constitutes improvement, which variables to manipulate, how to ensure fair testing, and how to justify their claims using collected data and physics principles. The task reflects a guided to open inquiry model, the authentic problem and constraints are provided (design an efficient safe lift system for a community building), but the investigative pathway, redesign, decision, performance metrics, and justification strategies are determined by students. In SPH3U, the inquiry leans closer to open inquiry as students independently construct mathematical models, identify controlled variables, analyze torque and work-energy relationships, and defend methodological limitations. In SPH4C, the inquiry remains authentic but is more guided, with more emphasis on reasoning and applied explanation. Across both contexts, the teacher acts as a facilitator and questioner rather than director, using formative assessment strategies to extend thinking rather than provide solutions.

The 5E instructional model is intentionally embedded throughout the four day sequence in a cohesive and iterative manner. **Engage** is represented through the initial elevator malfunction scenario and probing questions that activate prior knowledge about gears, force, motion, friction, and trade offs, as well as through repeated reframing of the challenge (baseline build, redesign focus, presentation rehearsal, and final presentation). Engagement is sustained by connecting the task to real world community infrastructure and by positioning students as designers responsible for safety and efficiency. **Explore** occurs primarily during the hands-on construction and testing phases, where students design prototypes, conduct baseline trials, manipulate components such as gear configurations and spool diameter, and collect both quantitative and qualitative performance data. Students encounter problems, frictional losses, and instability firsthand before formal explanation is required, ensuring that conceptual understanding emerges from evidence rather than lecture. **Explain** develops progressively during consolidation phases and intensifies on Day 3, when students must transform raw data into structured claims supported by Free Body diagrams, calculations, and physics reasoning. In SPH3U, this includes explicit connections to Newton's Laws, torque equations, and energy conservation; in SPH4C, it emphasizes clear cause and effect reasoning grounded in observed performance. Elaboration (**Elaborate)** is evident through the redesign cycle, where students apply baseline findings to refine their systems under controlled conditions, anticipate trade offs (speed vs stability, ease vs safety), and extend their understanding to real world elevator systems and community considerations. The presentation further elaborates understanding by requiring

transfer beyond experimentation into persuasive communication and applied feasibility. **Evaluate** is embedded continuously through assessment FOR, AS, and OF, teacher conferencing and questioning during exploration, peer reflection and metacognitive prompts during redesign, and the culminating presentation, rubric evaluation, Q&A defence, and peer/group evaluation on Day 4. Thus assessing both conceptual understanding and the integrity of the inquiry process itself.

The *Community Lift Design Challenge* exemplifies a rich assessment task (RAT) because it integrates multiple curriculum expectations within a sustained and authentic engineering inquiry. Students are not assessed on isolated skills, rather they must demonstrate understanding across the full investigative cycle, designing, testing, analyzing data, refining their system, and defending their conclusions using physics concepts and principles. Knowledge and application are inseparable, as students must apply concepts such as force interactions, torque, mechanical advantage, friction, and energy efficiency to improve performance and justify their decisions. Students then create a final presentation that requires students to communicate persuasively, support claims with data, acknowledge trade offs, and respond to critical questioning. This moves the task beyond model construction into disciplined, evidence based argumentation. Students must not only show their design world, but explain why it works using appropriate disciplinary reasoning.. Learning is documented and evaluated through prototype data collection (Handout 1, FOR), conceptual modelling via Free Body Diagrams (Handout 2, FOR), iterative redesign analysis, peer and self reflection (Handouts 3 and 4, AS), teacher conferencing and a culminating rubric that prioritizes Knowledge & Understanding and Application. Both process and product are assessed, and feedback is embedded throughout the inquiry cycle. Finally the task maintains difficulty while differentiating cognitive demand. SPH3U emphasizes abstract modelling and quantitative justification grounded in Newtonian mechanics, while SPH4C focuses on applied system optimization and functional reasoning. In both pathways, students engage in sustained inquiry, analytical decision making, and evidence based explanation. Thus this task exemplifies a RAT as it is integrative, authentic, inquiry driven, and intellectually demanding and puzzling.

Implementation Considerations: Student Responsibilities, Task Modifications, and Assessment

Across both courses, students are expected to work collaboratively as engineers and think critically as physicists. This means they must participate actively in the design, test, and redesign cycle, contribute meaningfully to group decision making , document evidence carefully in Handout 1, construct free Body Diagrams in Handout 2 with accuracy, and engage respectfully in peer reflection and evaluation through Handouts 3 and 4. All students are responsible for maintaining accurate data records, participating in fair testing procedures, and grounding claims in observable evidence rather than opinion. They are also expected to use appropriate physics vocabulary during discussion and presentations, ask clarifying questions, and reflect honestly on both the strengths and limitations of their design and collaboration process. However, expectations differ in depth and independence between the two courses.

In SPH3U, students are expected to demonstrate a higher degree of independence, abstraction, and precision. They must take greater ownership over identifying variables, designing controlled trials, and mathematically justifying claims using torque relationships, work energy analysis, and percent efficiency comparisons. Their Free Body Diagrams must be fully constructed without scaffolding, and their explanations must connect data to theory using formal physics reasoning. They are expected to critique methodological limitations and defend their reasoning during Q&A with minimal prompting. Thus, grade 11 students are expected to function with increasing academic autonomy and theoretical sophistication. Whereas in SPH4C, expectations emphasize applied reasoning, structured thinking, and practical justification. While students are still responsible for fair testing, data collection, and collaborative engagement, greater scaffolding is provided to support organization and conceptual clarity. Independence is still encouraged, but teacher check ins and formative conferencing play a stronger role in supporting clarity and confidence. The emphasis is on demonstrating understanding through application and clear explanation rather than abstract derivation.

Assessment strategies are deliberately layered and continuously across both courses. Assessment FOR learning (AFL) occurs through ongoing teacher observation, anecdotal notes, strategic questioning, mid investigation conferencing to clarify misconceptions, quick formative checks of Free body Diagrams (Handout 2), and review of data tables prior to retesting (Handout 1). These strategies allow the teacher to adjust instruction responsively and provide immediate descriptive feedback focused on reasoning quality rather than simple correctness. Assessment AS learning (AAL) is embedded through structured peer and self reflection in Handout 3, group discussions and evaluation in Handout 4, rehearsal feedback before the final presentation, and prompts requiring students to identify what evidence most strongly supports their claim. These opportunities foster metacognitive awareness and help students monitor their own growth in reasoning and collaboration. Assessment OF learning (AOF) culminates in the final presentation and reflective discussion, evaluated using a rubric with the strongest weight on Knowledge & Understanding and Application. The primary reason that all assessment components, including handouts and presentations, are to be completed on paper was to minimize the influence of AI in students' work. With AI tools becoming increasingly accessible and integrated into daily life, there was a concern that students might rely on these technologies to complete the tasks, rather than develop their own understanding. By using paper based assessments, we aimed to create opportunities for students to engage deeply with the material, think critically, and express their own ideas without the immediate availability of AI.

Throughout both courses, principles of Differentiated Instruction (DI) and Universal Design for Learning (UDL) ensure that expectations remain rigorous while access remains equitable (Ontario Ministry of Education, 2010). Multiple means of engagement are embedded through the authentic community elevator context, collaborative role selection, iterative improvement framing, and real world decision maker audience. Multiple means of representation are provided through video demonstrations, physical prototypes, teacher modelled diagrams, guided handouts, verbal discussions, visual data displays, and mathematical analysis. Multiple

means of action and expression allow students to demonstrate understanding through oral presentation, written explanation, diagrammatic modelling, quantitative analysis, and reflective evaluation. Within each class, flexible grouping and tiered questioning during consolidation, scaffolded supports (particularly in SPH4C), and extension opportunities (particularly in SPH3U) respond to varied readiness levels and learning profiles. Clear success criteria, descriptive feedback cycles, and structured reflection ensure that students can both demonstrate their strengths (hands on design, collaboration, applied reasoning) and develop identified areas of need (conceptual precision, analytical justification, communication clarity). Collectively, these expectations and assessment strategies create a rigorous, inquiry driven learning environment that balances high academic standards with responsive and inclusive instructional design across both Grade 11U and Grade 12C contexts.

Differentiation Between the Courses

In SPH3U (Grade 11 University Physics), the *Community Lift Design Challenge* is implemented as a rigorous theory driven engineering investigation that emphasises abstract modelling, mathematical analysis, and formal scientific justification. Students are expected to move beyond describing what worked and instead explain why their design performed as it did using Newton's Laws, torque relationships, mechanical advantage, and work-energy principles. Their Free body Diagrams must be complete and precise, including correctly labeled force interactions and system motion. Quantitative reasoning is central, students calculate percent improvement, compare input and output work, analyze efficiency, and justify trade offs using multi step algebraic reasoning. They are also expected to evaluate the validity of their experimental design by identifying controlled variables, potential sources of error, and reliability considerations. During the final presentation , students must defend their claims using formal physics vocabulary and respond to probing questions that require conceptual transfer and abstract reasoning. The emphasis in SPH3U is therefore on deep theoretical understanding, mathematical coherence, and disciplined scientific argumentation.

In SPH4C (Grade 12 College Physics), while the authentic engineering context and inquiry structure remain intact the task is intentionally modified to better align with students' learning strengths and needs, with significantly stronger emphasis on applied reasoning, structured support, and practical understanding. Mathematical abstraction is reduced, instead of multi step derivations or symbolic modelling, students focus on proportional reasoning, simplified efficiency comparisons, and clear quantitative explanations of how design changes affect performance. Free Body diagrams are scaffolded with partially structured templates that highlight the key forces required for their diagrams, guiding students in identifying forces without requiring the same level of analytical depth. The redesign phase would include more of a guided checkpoint style to ensure students intentionally control variables and understand what constitutes fair testing. Additionally, greater emphasis is placed on real world usability, safety considerations, system reliability, and feasibility in a community building context, rather than on theoretical modelling. Presentation expectations prioritize clarity of communication and practical

justification over abstract defence. These modifications preserve the integrity of the inquiry and the richness of the assessment task while ensuring that SPH4C students can demonstrate their strengths in hands-on problem solving and applied thinking, while receiving the scaffolding necessary to further develop their conceptual reasoning and analytical skills.

Assessment Rubric for the SPH3U Inquiry Task

The rubric for the SPH3U version of this inquiry task is provided at the end of the document alongside an external link to preserve the layout and readability. The rubric is designed to assess the key knowledge and skills students are expected to demonstrate throughout the task; specifically the presentation portion of this RAT. The rubric is designed to make the assessment criteria transparent to learners. In alignment with assignment expectations, the rubric clearly identifies what students will be assessed on and how performance will be evaluated, using a format that best fits the assessment demands of this inquiry-based task.

SPH3U was considered as the starting point for the rubric development as it sets the highest conceptual and communication demands for this task. This provided a clear baseline for the expectations that needed to remain consistent across both courses. An analytical rubric was selected because the presentation involves multiple distinct outcomes. Outcomes may include describing the system and design change, using evidence from testing, applying physics reasoning and calculations, and communicating these ideas clearly. Separating these criteria makes expectations explicit and supports specific, actionable feedback. Both rubrics were written to be flexible and accessible. The rubrics keep the same criteria and weighting, but allow for multiple entry points while providing opportunities for extension. This is important because students in both pathways can show a wide range of readiness, and some SPH3U students may still require applied-level supports. The SPH4C rubric therefore maintains the same overall structure as the SPH4U rubric. The SPH4C rubric differs primarily from the SPH3U rubric through simpler wording and more emphasis is placed towards applied reasoning and proportional comparisons. It also makes fair testing expectations more explicit and underlines practical design factors such as safety, reliability, and feasibility. Assessment is communicated to students using four achievement levels, a format consistent with Ontario assessment practices (Ontario Ministry of Education, 2010). Although each level on the rubric is associated with a percentage range, students are not assigned a specific percentage on the rubric. The percentage ranges are included for teacher tracking and reporting purposes, but feedback to students is communicated as an overall achievement level based on the preponderance of evidence across the criteria. This more holistic approach keeps the focus on strengths and next steps rather than point totals, while still meeting school requirements for summarizing achievement.

Link to the rubrics:

SPH3U: EDUC 4201/4224: IBL Task ZA & JD

SPH4C: EDUC 4201/4224: IBL Task ZA & JD

Link to resources for implementation of the task

Below are links to the documents used in this activity. Resources are submitted concurrently alongside this assignment. Resources were not merged with this document due to formatting and incompatibility issues (e.g., differing page orientations and file types). Although not required as per the outline and rubric, supplementary resources are included below as optional references to support lesson implementation and extension.

- Tracking Sheet
- Lesson Plans
- SPH3U Handout
- SPH4C Handout

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RUBRIC: SPH4C Community Design Challenge Presentation

NAME:
DATE:

Criteria	Level 4 100% - 80% (A+ to A-)	Level 3 79% - 70% (B+ to B-)	Level 2 69% - 60% (C+ to C-)	Level 1 59% - 50% (D+ to D-)	Level R < 50%
System Description and Design Change (Knowledge & Understanding) 30%	Clearly and accurately explains how the elevator system functions as a whole (input to transfer to output) Accurately explains the role of each major component (spool, gear, pulley, cable, load, etc.) Precisely identifies what specific component(s) were changed (e.g., gear, spool size, cable path, etc.) and explains their mechanical function Justifies the design change using explicit mechanical principles (e.g., torque, mechanical advantage, friction, efficiency) Uses accurate and specific terminology consistently	Provides an accurate explanation of overall system function though some details may be simplified Identifies the design change and explains its purpose Refers to relevant mechanical principles, though explanation may lack depth Describes how the change affected performance, but connections may be general Uses appropriate terminology with minor inconsistencies	Provides a partial or simplified explanation of how the system works Identifies the design change but justification is limited or unclear Makes general reference to mechanical ideas without clearly explaining them Limited discussion of how the change affected system performance Terminology may be inconsistent or occasionally incorrect	Explanation of system operation is unclear, fragmented, or contains misconceptions Design change is vaguely described or poorly justified Minimal connection to mechanical principles Terminology is largely incorrect or absent	System description and design change are missing or fundamentally incorrect Demonstrates little to no understanding of the mechanical system
Evidence from Testing (Thinking & Inquiry) 15%	Collects reliable and sufficient data from both baseline and modified trials Ensures a fair test by controlling variables (same load, height, procedure) Organizes data clearly in properly labeled tables and/or	Collects appropriate data from both trials Attempts fair testing with minor inconsistencies Data is organized and generally supports	Data is limited, incomplete, or inconsistently collected Fair testing procedures are inconsistently applied Evidence only partially supports conclusions	Data collection is minimal or poorly organized Comparison between trials is unclear or invalid Conclusions are weakly supported or unsupported by	Meaningful data is not provided No fair comparison between trials is evident

	graphs (with units) Uses evidence explicitly and logically to support claims about performance changes and how it benefits a real world system	conclusions Makes reference to evidence, though analysis may be surface level	Limited comparison between baseline and modified trials	evidence	
Physics Explanations and Calculations (Knowledge & Understanding) 15%	Correctly applies relevant physics ideas (for example force, friction, torque, work, efficiency) to explain why performance changed. Includes accurate calculations or clear quantitative comparisons that fit the task Shows steps clearly enough that someone else could follow. Clearly connects the numbers and the physics ideas to what happened during testing. Uses a force diagram/free body diagram (using a provided template if applicable) to identify key forces on the load, and connects it to the explanation.	Applies relevant physics concepts with minor errors Calculations are mostly correct but maybe forgetting a few units or steps Makes general connections between physics and observed results Explanation may rely more on formula use than conceptual depth	Physics concepts are partially correct or inconsistently applied Calculations may contain errors or incomplete reasoning Limited or unclear connection between physics and system performance	Physics explanations are largely incorrect or missing Calculations are incorrect or absent	No meaningful physics explanation or calculation is provided
Application & Design Effectiveness (Application) 30%	Design change results in a clear and measurable improvement (efficiency, control, force required, smoothness). Design decisions are informed by test evidence and physics reasoning.	Design shows measurable improvement supported by evidence Makes relevant real world connections	Design shows limited effectiveness or inconsistent improvement Evidence is limited or incomplete	Design change shows little improvement or is not justified Limited or inaccurate application of physics	No meaningful applications of learning is demonstrated

	Explains the design using real-world context with emphasis on usability (safety, reliability, feasibility for a community lift). References are provided to support/validate real world application information Identifies at least one limitation and a realistic next improvement.	Physics reasoning is present but may lack depth Limited evaluation of design limitations	Real world connections are vague or underdeveloped Minimal evaluation of design performance	Minimal connection to real world systems	
Presentation (Communication) 5%	Presentation is clear, logical, and well structured Visuals (diagrams, graphs) are accurate, labeled, and directly enhance understanding All group members contribute meaningfully demonstrating shared responsibility Responses to questions demonstrate understanding of the project	Presentation is organized and understandable Visuals support understanding but may lack precision Most group members participated or participation was not fairly shared Responses to questions show general understanding	Presentation lacks clarity or organization Visuals are limited or unclear Participation is uneven Responses to questions show limited understanding	Presentation is difficult to follow Minimal to no contribution from some group members Limited ability to answer questions	Presentation is incomplete or not delivered

Grade Level:_____

Additional Comments:

Strengths:

Area of Improvement:

RUBRIC: SPH3U Community Design Challenge Presentation

NAME:
DATE:

Criteria	Level 4 100% - 80% (A+ to A-)	Level 3 79% - 70% (B+ to B-)	Level 2 69% - 60% (C+ to C-)	Level 1 59% - 50% (D+ to D-)	Level R < 50%
System Description and Design Change (Knowledge & Understanding) 30%	Clearly and accurately explains how the elevator system functions as a whole (input to transfer to output) Accurately explains the role of each major component (spool, gear, pulley, cable, load, etc.) Precisely identifies what specific component(s) were changed (e.g., gear, spool size, cable path, etc.) and explains their mechanical function Justifies the design change using explicit mechanical principles (e.g., torque, mechanical advantage, friction, efficiency) Uses accurate and specific terminology consistently	Provides an accurate explanation of overall system function though some details may be simplified Identifies the design change and explains its purpose Refers to relevant mechanical principles, though explanation may lack depth Describes how the change affected performance, but connections may be general Uses appropriate terminology with minor inconsistencies	Provides a partial or simplified explanation of how the system works Identifies the design change but justification is limited or unclear Makes general reference to mechanical ideas without clearly explaining them Limited discussion of how the change affected system performance Terminology may be inconsistent or occasionally incorrect	Explanation of system operation is unclear, fragmented, or contains misconceptions Design change is vaguely described or poorly justified Minimal connection to mechanical principles Terminology is largely incorrect or absent	System description and design change are missing or fundamentally incorrect Demonstrates little to no understanding of the mechanical system
Evidence from Testing (Thinking & Inquiry) 15%	Collects reliable and sufficient data from both baseline and modified trials Ensures a fair test by controlling variables (same load, height, procedure) Organizes data clearly in properly labeled tables and/or	Collects appropriate data from both trials Attempts fair testing with minor inconsistencies Data is organized and generally supports	Data is limited, incomplete, or inconsistently collected Fair testing procedures are inconsistently applied Evidence only partially supports conclusions	Data collection is minimal or poorly organized Comparison between trials is unclear or invalid Conclusions are weakly supported or unsupported by	Meaningful data is not provided No fair comparison between trials is evident

	graphs (with units) Uses evidence explicitly and logically to support claims about performance changes	conclusions Makes reference to evidence, though analysis may be surface level	Limited comparison between baseline and modified trials	evidence	
Physics Explanations and Calculations (Knowledge & Understanding) 15%	Correctly applies relevant physics concepts (e.g., torque, work, efficiency) Includes accurate calculations with formulas, units, and logical steps clearly shown Explicitly links calculations and physics principles to observed system preferences Demonstrates conceptual understanding beyond formula substitution (explaining why the change affected performance)	Applies relevant physics concepts with minor errors Calculations are mostly correct but may omit units or steps Makes general connections between physics and observed results Explanation may rely more on formula use than conceptual depth	Physics concepts are partially correct or inconsistently applied Calculations may contain errors or incomplete reasoning Limited or unclear connection between physics and system performance	Physics explanations are largely incorrect or missing Calculations are incorrect or absent	No meaningful physics explanation or calculation is provided
Application & Design Effectiveness (Application) 30%	Design modification(s) results in a clear and measurable improvement (efficiency, control, force required, smoothness) Design decisions are directly informed by evidence and physics reasoning Makes clear connections to real world mechanical systems (elevators, cranes, lifting devices) References are provided to support/validate information	Design shows measurable improvement supported by evidence Makes relevant real world connections Physics reasoning is present but may lack depth Limited evaluation of design limitations	Design shows limited effectiveness or inconsistent improvement Evidence is limited or incomplete Real world connections are vague or underdeveloped Minimal evaluation of design performance	Design change shows little improvement or is not justified Limited or inaccurate application of physics Minimal connection to real world systems	No meaningful applications of learning is demonstrated

Presentation (Communication) 5%	Presentation is clear, logical, and well structured Visuals (diagrams, graphs) are accurate, labeled, and directly enhance understanding All group members contribute meaningfully demonstrating shared responsibility Responses to questions demonstrate understanding of the project	Presentation is organized and understandable Visuals support understanding but may lack precision Most group members participated or participation was not fairly shared Responses to questions show general understanding	Presentation lacks clarity or organization Visuals are limited or unclear Participation is uneven Responses to questions show limited understanding	Presentation is difficult to follow Minimal to no contribution from some group members Limited ability to answer questions	Presentation is incomplete or not delivered

Grade Level:_____

Additional Comments:

Strengths:

Area of Improvement: